



DISCOVERY 2K26

ROBO SOCCER RULES AND REGULATIONS

1. EVENT OVERVIEW

DISCOVERY 2K26 – Robo Soccer is a team-based robotic football competition designed to test robot control, strategy, coordination, attacking ability, defensive planning, and technical skills.

The competition will be conducted in a **2 vs 2 robot format**, consisting of:

- **One wired defensive robot** provided by the organizing committee.
- **One wireless robot** operated by the participating team.
- The wired robot is positioned at the team's own goal for defensive purposes.
- The wireless robot is the active robot and is allowed to perform both attacking and defensive movements.

2. TEAM STRUCTURE

- Each team shall consist of **2 participants**.
- Each team will have **2 robots**:
 1. One **wired defensive robot**, provided by the organizing committee.
 2. One **wireless robot**, which may be brought by the participating team or obtained through the organizing committee, subject to approval.
- The two participants may operate/control the permitted robots according to the event instructions.
- Teams must follow the robot configuration and technical requirements specified by the organizing committee.

3. ROBOT CONFIGURATION

Each team shall have the following configuration:

Robot 1 – Wired Defensive Robot

- The wired defensive robot will be **provided by the organizing committee**.
- It will be positioned near the team's own goal post.
- Its primary purpose is **defending the goal**.
- The wired robot **must not cross the centre line**.
- The wired robot must remain within its permitted defensive half.
- The robot and its wired control system will be supplied by the organizing committee.
- Participants are not permitted to modify the provided wired robot.

Robot 2 – Wireless Robot

- The wireless robot may be:
 - brought by the participating team, **or**

- arranged/provided by the organizing committee, subject to availability and approval.
- The wireless robot is the team's **active field robot**.
- It may perform both **attack and defence**.
- The wireless robot is permitted to cross the centre line.
- The wireless robot may move throughout the permitted playing area.
- The wireless robot must comply with all applicable event safety and technical rules.

Robot Configuration Assistance

Teams requiring clarification or assistance regarding the wireless robot configuration should contact the **organizing committee/management committee** before the match.

The organizing committee will have the final authority regarding whether a robot configuration is acceptable.

4. PLAYING FIELD / TRACK



The official playing track dimensions are:

- **Track Length:** 200 cm
- **Track Width:** 180 cm

Therefore, the playing area will be:

200 cm × 180 cm

The field will be divided into two halves by a clearly marked **centre line**.

Centre Line Rule

- The centre line will divide the field into the two team halves.
- The **wired defensive robot must not cross the centre line**.
- The wireless robot is permitted to cross the centre line.
- The wireless robot may enter the opponent's half for attacking.
- The wireless robot may return to its own half for defensive play.

5. GOAL POST

- Each team will have one goal post.
- **Goal width: 40 cm**
- The goal will be positioned at the respective end of the playing field.
- The goal area will be clearly marked.
- A goal will be awarded when the ball completely crosses the goal line.

The exact goal height/depth, if applicable, will be specified by the organizing committee before the event.

6. BALL

- A **plastic table-tennis ball** will be used.
- The same approved ball will be used during the competition.
- Participants are not permitted to replace or modify the official ball.
- The referee may replace the ball if it becomes damaged or unsuitable for play.

7. MATCH FORMAT

- The match will be played between **two teams**.
- Each team will have:
 - One wired defensive robot.
 - One wireless active robot.
- Therefore, **four robots may be associated with the match**, with one wired defensive robot positioned at each team's goal area and one wireless robot from each team operating on the field.
- The main objective is to score more goals than the opponent within the allotted time.

8. MATCH DURATION

- Each match will be **3 minutes** long.
- The match will start after the referee gives the start signal.
- The ball will initially be placed at the centre of the field.
- The match will continue until the end of the allotted time unless stopped by the referee.
- The referee may stop the match for safety, technical problems, or other valid reasons.

9. KICK-OFF

- The ball will be placed at the centre of the field before the start.
- Both wireless robots must remain in their designated starting positions until the referee gives the start signal.
- The wired defensive robots will remain at their respective goal areas.
- No participant may move a robot before the official start signal.
- After a goal, the ball will be returned to the centre and play will restart on the referee's signal.

10. WIRED DEFENSIVE ROBOT RULES

The wired defensive robot is a special organizer-provided robot intended to defend the team's goal.

Mandatory Rules

- The wired defensive robot must remain on its own side of the centre line.
- **Crossing the centre line is strictly prohibited.**

- The robot must remain within its designated defensive area.
- The robot may be used to block, deflect, or intercept the ball.
- The wired robot must not intentionally damage the opponent's wireless robot.
- The wire must remain safely positioned and must not be intentionally used to obstruct the opponent.
- Participants must not pull, throw, or deliberately manipulate the wire to interfere with the opponent.

11. WIRELESS ROBOT RULES

The wireless robot is the primary active robot of each team.

The wireless robot:

- May cross the centre line.
- May enter the opponent's half.
- May attack the opponent's goal.
- May return to its own half.
- May assist in defending its own goal.
- May push, strike, or deflect the ball.
- Must not intentionally damage the opponent's robot.
- Must not intentionally trap or permanently hold the ball.

12. WIRELESS ROBOT CONFIGURATION

A team may use:

Option 1 – Own Robot

A team may bring its own wireless robot, provided that it meets the technical and safety requirements of the event.

Option 2 – Organizer-Provided Robot

A team may request a wireless robot from the organizing committee, subject to availability.

Approval

- Teams intending to use their own robot should contact the organizing committee before the event.
- The organizing committee may inspect the robot.
- The robot may be rejected if it is unsafe or does not comply with the event requirements.
- The decision of the organizing committee regarding robot eligibility will be final.

13. ROBOT TECHNICAL REQUIREMENTS

For an externally supplied wireless robot, the following requirements apply unless otherwise approved by the organizing committee:

- Maximum allowable weight: **3 kg**
- Maximum number of motors: **4 motors**
- Maximum motor speed: **1000 RPM per motor**
- Manual control is compulsory.
- Wireless control is compulsory for the active wireless robot.
- Wired control of the active wireless robot is prohibited.
- All wires must be properly insulated and securely arranged.
- The robot must be safe for operation on the playing field.

The organizing committee may conduct a technical inspection before the match.

14. PROHIBITED SYSTEMS

The following systems are prohibited:

- Pneumatic systems
- Hydraulic systems
- Combustion systems
- Flammable liquids
- Explosive mechanisms
- RF jammers
- Devices designed to interfere with another robot's wireless control
- Dangerous electrical systems
- Any mechanism designed to intentionally damage another robot
- Any unauthorized mechanism that creates a safety hazard

Violation of these rules may result in immediate disqualification.

15. BALL CONTROL

The wireless robot may:

- Push the ball.
- Strike the ball.
- Deflect the ball.
- Intercept the ball.
- Move the ball toward the opponent's goal.

The robot must not:

- Permanently trap the ball.
- Carry the ball in an illegal manner.
- Enclose the ball to prevent an opponent from accessing it.
- Use adhesive or similar mechanisms to retain the ball.

The referee shall have the final authority to determine whether ball control is legal.

16. GOAL SCORING

A goal will be counted when the **entire ball crosses the opponent's goal line.**

After a goal:

1. The referee will confirm the goal.
2. The score will be recorded.
3. The ball will be returned to the centre.
4. The robots will return to their designated positions.
5. Play will restart on the referee's signal.

17. FOULS

The following actions may be considered fouls:

1. Wired defensive robot crossing the centre line.
2. Intentional damage to another robot.
3. Aggressive or unsafe collisions.
4. Intentional obstruction.
5. Illegal ball trapping.
6. Interference with another team's wireless control.

7. Unauthorized modification of the organizer-provided wired robot.
8. Use of prohibited systems.
9. Deliberate misuse of the wired robot's cable.
10. Unsafe operation of a robot.
11. Misconduct toward referees or other participants.

18. PENALTIES

Depending on the severity of the violation, the referee may impose:

- Verbal warning.
- Official warning.
- Temporary stoppage of the match.
- Awarding a goal to the opposing team.
- Removal of the offending robot from play.
- Disqualification of the team.

Repeated or serious violations may result in immediate disqualification.

19. WIRED ROBOT CABLE RULES

Because the defensive robot is wired:

- The cable must be handled safely.
- The cable must not be deliberately used to obstruct an opponent.
- Participants must not pull the cable aggressively.
- Participants must not wrap the cable around another robot.
- Participants must not intentionally place the cable across the opponent's path.
- The cable must remain connected to the provided control system.
- Only authorized organizers may repair or modify the wired robot or its cable.

Any deliberate misuse of the cable may be treated as a foul.

20. ROBOT FAILURE

Wired Robot Failure

If the organizer-provided wired defensive robot fails:

- The participant must immediately inform the referee.
- The technical team will inspect the robot.
- The organizing committee may repair or replace the robot.
- The referee may restart or continue the match depending on the circumstances.

Wireless Robot Failure

If a team's wireless robot fails:

- The team must inform the referee.
- No participant may enter the field without permission.
- Repair or replacement will be allowed only according to the decision of the organizing committee.
- A replacement robot must satisfy the event's technical requirements.

21. SAFETY RULES

- All participants must follow safety instructions.
- Robots must not have dangerous exposed components.
- Sharp or hazardous projections are prohibited.

- Electrical connections must be properly insulated.
 - Participants must not intentionally drive robots toward people.
 - Only authorized persons may enter the playing area during a match.
 - Participants must not touch a moving robot unless instructed by the referee.
 - The referee may stop the match immediately if a safety hazard occurs.
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22. TIE-BREAKER

If both teams have the same score at the end of the 3-minute match:

- A tie-break procedure will be conducted.
- The referee will conduct a toss to determine the first attacking team.
- The wireless robot will be used for the attacking attempt.
- The wired defensive robot will remain at its designated goal position.
- The attacking wireless robot will have a maximum of **10 seconds** to score.
- If the attacking team scores, it wins the attempt.
- If the attacking team fails to score within 10 seconds, the opposing team wins the attempt.
- The referee's decision will be final.

If required by the organizing committee, additional tie-break attempts may be conducted.

23. PARTICIPANT RESPONSIBILITIES

Each participant must:

- Report to the event area on time.
 - Complete robot inspection before the match.
 - Use the assigned organizer-provided wired robot.
 - Use an approved wireless robot.
 - Follow all referee instructions.
 - Operate the robot responsibly.
 - Avoid intentional collisions or damage.
 - Maintain sportsmanship.
 - Immediately report technical problems.
 - Not modify organizer-provided equipment without permission.
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24. MISCONDUCT

The following behavior may result in disciplinary action:

- Abusive language.
- Aggressive behavior.
- Disrespect toward referees or organizers.
- Intentional equipment damage.
- Cheating.
- Interference with another team's wireless signal.
- Unauthorized robot modification.
- Deliberate violation of the centre-line restriction.
- Repeated violation of safety rules.

The organizing committee may disqualify a team for serious misconduct.

25. REFEREE AUTHORITY

The referee has the authority to:

- Start and stop matches.
- Confirm goals.
- Call fouls.
- Apply penalties.
- Stop play for safety reasons.
- Determine whether a robot has violated the centre-line rule.
- Determine whether a robot action is legal.
- Request technical inspection.
- Restart a match when necessary.

The referee's decision during the match shall be final and binding.

26. ORGANIZING COMMITTEE AUTHORITY

The organizing committee reserves the right to:

- Inspect robots.
- Approve or reject participant-provided wireless robots.
- Provide wireless robots when requested and available.
- Repair or replace organizer-provided robots when necessary.
- Modify match arrangements for safety or technical reasons.
- Make reasonable changes to the competition procedure.
- Take final decisions regarding technical disputes and equipment.

Any major rule changes will be communicated to participating teams.

27. FIELD DIMENSIONS – FINAL

Parameter	Specification
Track Length	200 cm
Track Width	180 cm
Playing Area	200 cm × 180 cm
Goal Width	40 cm
Centre Line	Marked at the centre of the track
Wired Robot	Defensive only
Wireless Robot	Attack + Defence
Wired Robot Centre-Line Crossing	Not allowed
Wireless Robot Centre-Line Crossing	Allowed

28. QUICK RULE

Parameter	Rule
Event	DISCOVERY 2K26 – Robo Soccer
Team Size	2 participants
Match Format	2 vs 2
Wired Robot	1 per team
Wired Robot	Provided by organizers
Wired Robot Role	Goal defence
Wired Robot Centre Line	Cannot cross
Wireless Robot	1 per team
Wireless Robot	Own or organizer-provided

Wireless Robot Role	Attack + Defence
Track Length	200 cm
Track Width	180 cm
Goal Width	40 cm
Match Duration	3 minutes
Ball	Plastic table-tennis ball
Wireless Robot Control	Wireless
Wired Robot Control	Organizer-provided wired control
Max Weight*	4 kg
Max Motors*	4
Max Motor Speed*	1000 RPM
RF Jamming	Prohibited
Tie-Break	10-second attack
Referee Decision	Final and binding

*Applicable to participant-provided wireless robots unless otherwise specified/approved by the organizing committee.

FINAL NOTE

All participants are required to read and understand the rules before participating in the competition.

By registering for the event, participants agree to follow the rules, safety requirements, referee instructions, and decisions of the organizing committee.

The organizing committee reserves the right to take the final decision on any situation not specifically covered in this rule book.

One important point: I have treated the **wired robot as a goal defender that cannot cross the centre line**, while the **wireless robot can freely attack and defend on both halves**.